

A Human-In-The-Loop Simulation Framework for Evaluating Control Strategies in Gait Assistive Robots

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1 Introduction

- As balance and mobility decline with age, fall risks rise—making gait assistive robots increasingly vital.
- Varied impairments and user behaviors make personalized control essential, but real-world testing is limited by ethical and practical constraints.
- Most HITL simulations model either separate or tightly coupled systems, overlooking the passive, responsive support needed in gait assistive robots.
- Gait assistive robots intervene only when needed—like during balance loss or fatigue—to minimize disruption to natural gait.

Goal: To develop a Human-In-The-Loop simulation framework specifically for gait assistive robots, to enable realistic evaluation and personalization of robot control strategy in a safe and rich environment.

2 Human-In-The-Loop Simulation Framework

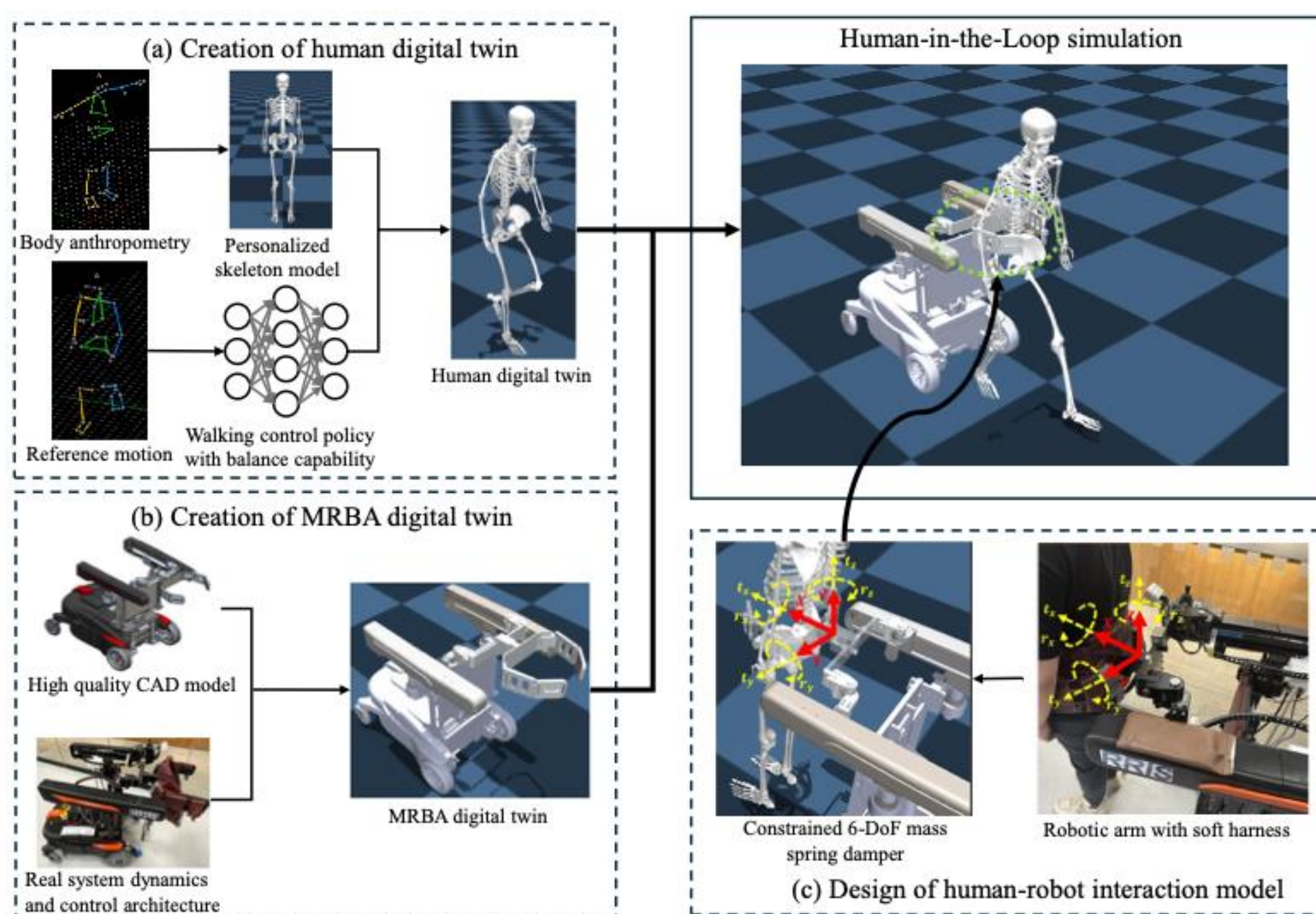


Figure 1: Overview of the Human-in-the-Loop (HITL) simulation framework for gait assistive robots. (a) Human digital twin: The process includes using body anthropometry to generate a personalized skeleton model. This model has a walking control policy with varied balance abilities based on the reference motion to create a human digital twin. (b) MRBA digital twin: A CAD model, combined with real system dynamics and control architecture, is used to develop the MRBA digital twin. (c) Physical Human-Robot Interaction (pHRI) model: The human digital twin is constrained with a six-DoF mass-spring-damper mechanism representing the robotic arm with a soft harness.

3 Experiments

- Healthy 28 year old male (95kg, 1.81m)
- Subject tightly attached to the robot using the onboard harness.
- Subject to walk naturally at their preferred speed for 10 meters, for 4 times
- Two controllers implemented and compared. Baseline PID control and a new personalized speed-adaptive controller [1].
- Motion data recorded with a markless mocap system [2].



Figure 2: Experiment setup. The subject was tightly attached to the robot via the onboard harness. Sufficient time was given to allow the subject to become comfortable walking with the robot under both two control conditions.

4 Results

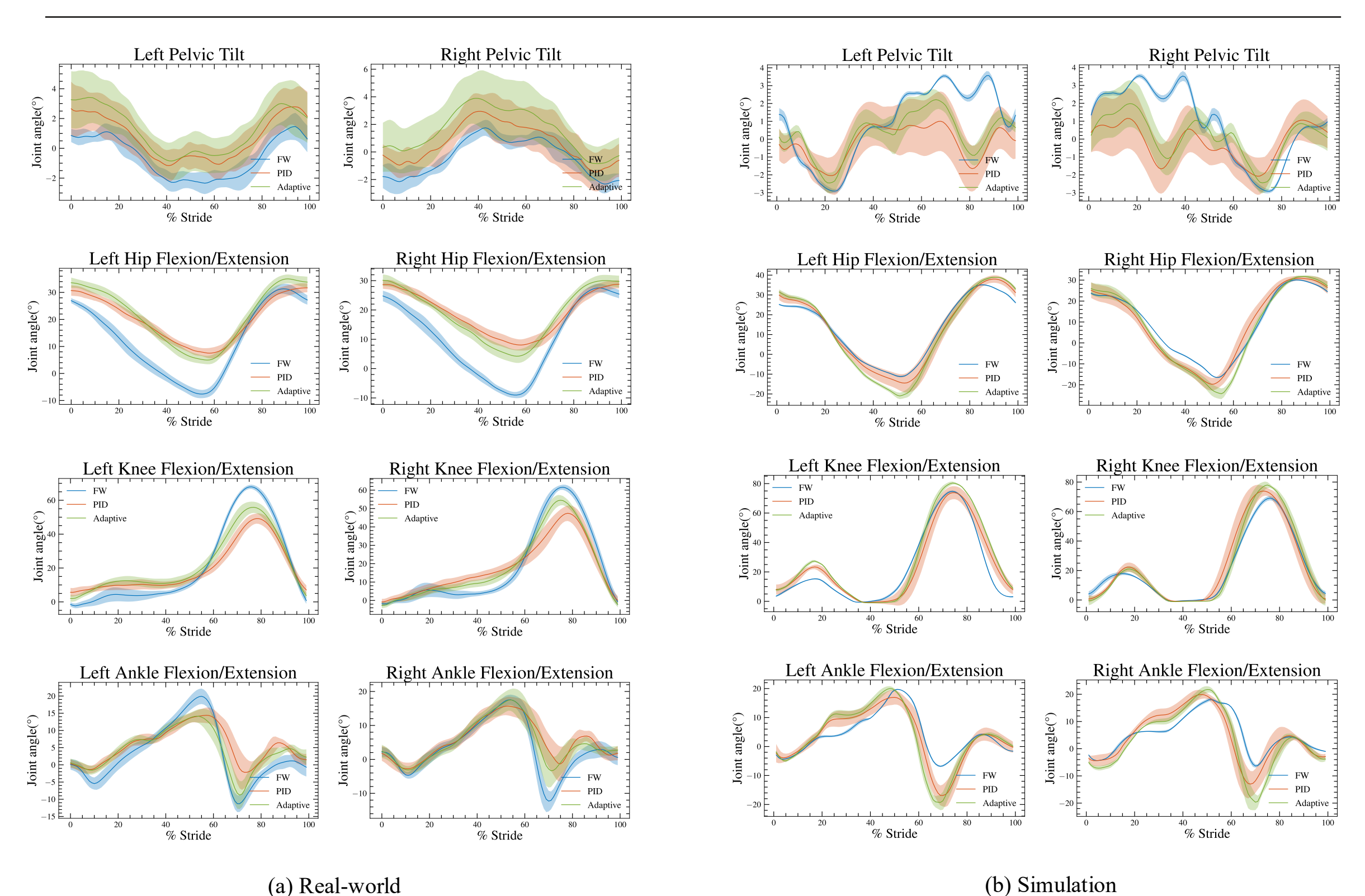


Figure 3: Comparison of lower limb joint kinematics for (a) real-world and (b) simulation. The graphs show the joint angles in the sagittal plane for the pelvis, hip, knee, and ankle joints over a complete gait cycle. FW indicates free walking without the robot, PID denotes the use of the PID controller for the follow-me function of MRBA, and Adaptive denotes the speed-adaptive controller for the follow-me function of MRBA. The shaded areas represent the standard deviation of the joint angles across multiple gait cycles.

COMPARISON OF INTER-CYCLE STANDARD DEVIATION OF JOINT ANGLES.

Task	Hip(°)		Knee(°)		Ankle(°)	
	Real	Sim	Real	Sim	Real	Sim
PID	1.87	2.96	3.16	4.43	2.28	2.67
Adaptive	1.82	1.61	2.55	2.11	2.20	1.45

COMPARISON OF SPATIOTEMPORAL GAIT PARAMETERS. THE VALUES ARE PRESENTED AS MEAN ± STANDARD DEVIATION

Task	Stride length (m)		Gait speed (m/s)	
	Real	Sim	Real	Sim
Free walking	1.29±0.04	1.17±0.01	1.04±0.02	1.12±0.01
PID	0.90±0.09	1.13±0.06	0.59±0.06	1.28±0.12
Adaptive	1.14±0.07	1.28±0.06	0.98±0.02	1.47±0.07

COMPARISON OF FORWARD (e_x) AND LATERAL (e_y) TRACKING ERRORS

Task	Avg. $ e_x $ (cm)		Avg. $ e_y $ (cm)	
	Real	Sim	Real	Sim
PID	4.25	7.65	3.32	7.73
Adaptive	0.37	0.7	0.72	0.53

- Adaptive controller outperforms PID controller in both real and simulated settings, achieving lower forward and lateral tracking errors, while better preserving natural gait dynamics

5 Conclusion & Future Work

- Both simulation and real-world experiments showed that the speed-adaptive controller improved compliance and reduced gait distortion.
- Our findings demonstrate the potential of our HITL framework to optimize control strategies for more responsive and transparent human-robot interaction.
- The human digital twin exhibited unexpected adaptations, offering valuable insights into simulating physical human-robot interaction.
- We plan to incorporate soft body and strap modeling to improve the realism of physical interactions, enhancing fidelity and enabling more personalized robot control.

References:

- [1] Y. Wang, M. Yuan, L. Li, K. S. G. Chua, W. T. Ang, et al., "Graceful user following for mobile balance assistive robot in daily activities assistance," IFAC-PapersOnLine, vol. 56, no. 2, pp. 1139–1144, 2023.
- [2] P. Jatesiktat, G. M. Lim, W. S. Lim, and W. T. Ang, "Anatomicalmarker-driven 3d markerless human motion capture," IEEE Journal of Biomedical and Health Informatics, 2024.

Paper

Video

